

Yoon Kim

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Employment

Massachusetts Institute of Technology

Jul 2025 – Present	Associate Professor
Jul 2021 – Jun 2025	Assistant Professor

Education

2020	Ph.D., Computer Science Harvard University Advisor: Alexander Rush Thesis: “Deep Latent Variable Models of Natural Language”
2015	M.S., Data Science New York University Advisor: David Sontag
2012	M.A., Statistics Columbia University
2009	B.A., Mathematics B.A., Economics Cornell University

Honors and Awards

2025	NSF CAREER Award
2025	Google ML and Systems Junior Faculty Award
2024	Schmidt Sciences AI2050 Early Career Fellowship
2020	Harvard University Award for Outstanding Doctoral Dissertation in CS
2018	Google Ph.D. Fellowship
2017	ACL Best Demo Paper Award Runner-Up
2014	KDD Cup 1st place
2013	NYU Full Merit Scholarship
2009	Cornell University Phi Beta Kappa

Past Employment

Jun 2019 – Jul 2021	MIT-IBM Watson AI Lab, Postdoctoral Researcher
May 2019 – Aug 2019	Google, Research Intern
May 2018 – Aug 2018	DeepMind, Research Intern
May 2017 – Aug 2017	Facebook, Research Intern
Jul 2012 – Aug 2016	American International Group, Data Scientist
Oct 2011 – Jul 2012	Barclays Capital, Analyst
Jul 2009 – Oct 2011	Morgan Stanley, Analyst
May 2008 – Aug 2008	Merill Lynch, Analyst Intern

Conference Publications

- NeurIPS '24 Songlin Yang, Bailin Wang, Yu Zhang, Yikang Shen, Yoon Kim. "Parallelizing Linear Transformers with the Delta Rule over Sequence Length." In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*.
- EMNLP '24 Zhaofeng Wu, Ananth Balashankar, Yoon Kim, Jacob Eisenstein, Ahmad Beirami. "Reuse Your Rewards: Reward Model Transfer for Zero-Shot Cross-Lingual Alignment." In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*.
- EMNLP '24 Yung-Sung Chuang, Linlu Qiu, Cheng-Yu Hsieh, Ranjay Krishna, Yoon Kim, James Glass. "Lookback Lens: Detecting and Mitigating Contextual Hallucinations in Large Language Models Using Only Attention Maps." In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*.
- EMNLP '24 Dong Won Lee, Hae Won Park, Yoon Kim, Cynthia Breazeal, Louis-Philippe Morency. "Improving Dialogue Agents by Decomposing One Global Explicit Annotation with Local Implicit Multimodal Feedback." In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*.
- EMNLP '24 Han Guo, William Brandon, Radostin Cholakov, Jonathan Ragan-Kelley, Eric P. Xing, Yoon Kim. "Fast Matrix Multiplications for Lookup Table-Quantized LLMs." In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (Findings)*.
- EMNLP '24 Peter Baile Chen, Yi Zhang, Chunwei Liu, Sejal Gupta, Yoon Kim, Michael Cafarella. "MDCR: A Dataset for Multi-Document Conditional Reasoning." In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (Findings)*.
- UIST '24 Joshua Gorniak, Yoon Kim, Donglai Wei, Nam Wook Kim. "VizAbility: Enhancing Chart Accessibility with LLM-based Conversational Interaction." In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*.
- COLM '24 Lucas Torroba Hennigen, Shannon Shen, Aniruddha Nrusimha, Bernhard Gapp, David Sontag, Yoon Kim. "Towards Verifiable Text Generation with Symbolic References." In *Proceedings of the 1st Conference on Language Modeling*.
- COLM '24 Lucas Torroba Hennigen, Shannon Shen, Aniruddha Nrusimha, Bernhard Gapp, David Sontag, Yoon Kim. "Towards Verifiable Text Generation with Symbolic References." In *Proceedings of the 1st Conference on Language Modeling*.
- COLM '24 Jillian Ross, Yoon Kim, Andrew W. Lo. "LLM economicus? Mapping the Behavioral Biases of LLMs via Utility Theory." In *Proceedings of the 1st Conference on Language Modeling*.
- ICML '24 Songlin Yang, Bailin Wang, Yikang Shen, Rameswar Panda, Yoon Kim. "Gated Linear Attention Transformers with Hardware-Efficient Training." In *Proceedings of the 41st International Conference on Machine Learning*.
- ICML '24 Ekin Akyürek, Bailin Wang, Yoon Kim, Jacob Andreas. "In-Context Language Learning: Architectures and Algorithms." In *Proceedings of the 41st International Conference on Machine Learning*.
- ICML '24 Yao Fu, Rameswar Panda, Xinyao Niu, Xiang Yue, Hannaneh Hajishirzi, Yoon Kim, Hao Peng. "Data Engineering for Scaling Language Models to 128K Context." In *Proceedings of the 41st International Conference on Machine Learning*.
- ACL '24 Shannon Zejiang Shen, Hunter Lang, Bailin Wang, Yoon Kim, David Sontag. "Learning to Decode Collaboratively with Multiple Language Models." In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*.

- ACL '24 Jerry Ngo, Yoon Kim. "What Do Language Models Hear? Probing for Auditory Representations in Language Models." In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*.
- ACL '24 William Merrill, Zhaofeng Wu, Norihito Naka, Yoon Kim, Tal Linzen. "Can You Learn Semantics Through Next-Word Prediction? The Case of Entailment." In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Findings)*.
- ACL '24 Yujun Mao, Yoon Kim, Yilun Zhou. "CHAMP: A Competition-level Dataset for Fine-Grained Analyses of LLMs' Mathematical Reasoning Capabilities." In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Findings)*.
- ACL '24 Hang Jiang, Xiajie Zhang, Robert Mahari, Daniel Kessler, Eric Ma, Tal August, Irene Li, Alex 'Sandy' Pentland, Yoon Kim, Jad Kabbara, Deb Roy. "Leveraging Large Language Models for Learning Complex Legal Concepts through Storytelling." In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*.
- NAACL '24 Zhaofeng Wu, Linlu Qiu, Alexis Ross, Ekin Akyürek, Boyuan Chen, Bailin Wang, Najoung Kim, Jacob Andreas, Yoon Kim. "Reasoning or Reciting? Exploring the Capabilities and Limitations of Language Models Through Counterfactual Tasks." In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics*.
- NAACL '24 Bowen Pan, Rameswar Panda, SouYoung Jin, Rogerio Feris, Aude Oliva, Phillip Isola, Yoon Kim. "LangNav: Language as a Perceptual Representation for Navigation." In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (Findings)*.
- NAACL '24 Tianhua Zhang, Jiaxin Ge, Hongyin Luo, Yung-Sung Chuang, Mingye Gao, Yuan Gong, Xixin Wu, Yoon Kim, Helen Meng, James Glass. "Natural Language Embedded Programs for Hybrid Language Symbolic Reasoning." In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (Findings)*.
- NAACL '24 Yue Zhou, Yada Zhu, Diego Antognini, Yoon Kim, Yang Zhang. "Paraphrase and Solve: Exploring and Exploiting the Impact of Surface Form on Mathematical Reasoning in Large Language Models." In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics*.
- ICLR '24 Han Guo, Philip Greengard, Eric P. Xing, Yoon Kim. "LQ-LoRA: Low-rank Plus Quantized Matrix Decomposition for Efficient Language Model Finetuning." In *Proceedings of the Twelfth International Conference on Learning Representations*.
- ICLR '24 Linlu Qiu, Liwei Jiang, Ximing Lu, Melanie Sclar, Valentina Pyatkin, Chandra Bhagavatula, Bailin Wang, Yoon Kim, Yejin Choi, Nouha Dziri, Xiang Ren. "Phenomenal Yet Puzzling: Testing Inductive Reasoning Capabilities of Language Models with Hypothesis Refinement." In *Proceedings of the Twelfth International Conference on Learning Representations*.
- ICLR '24 Yung-Sung Chuang, Yujia Xie, Hongyin Luo, Yoon Kim, James Glass, Pengcheng He. "DoLa: Decoding by Contrasting Layers Improves Factuality in Large Language Models." In *Proceedings of the Twelfth International Conference on Learning Representations*.
- CoNLL '23 Shinjini Ghosh, Yoon Kim, Ramon Fernandez Astudillo, Tahira Naseem, Jacob Andreas. "Alignment via Mutual Information." In *Proceedings of the 2023 The SIGNLL Conference on Computational Natural Language Learning*.
- EMNLP '23 Zilu Tang, Mayank Agarwal, Alex Shypula, Bailin Wang, Derry Wijaya, Jie Chen, Yoon Kim. "Explain-then-Translate: An Analysis on Improving Program Translation with Self-generated Explanations." In *Proceedings of the 2023 Conference on Empirical Methods in Nat-*

- EMNLP '23 Wei Liu, Songlin Yang, Yoon Kim, Kewei Tu. "Simple Hardware-Efficient PCFGs with Independent Left and Right Productions." In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (Findings)*.
- EMNLP '23 Hongyin Luo, Yung-Sung Chuang, Yuan Gong, Tianhua Zhang, Yoon Kim, Xixin Wu, Danny Fox, Helen Meng, James Glass. "SAIL: Search-Augmented Instruction Learning." In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (Findings)*.
- NeurIPS '23 Bailin Wang, Zi Wang, Xuezhi Wang, Yuan Cao, Rif A. Saurous, Yoon Kim. "Grammar Prompting for Domain-Specific Language Generation with Large Language Models." In *Proceedings of the 35th Conference on Neural Information Processing Systems*.
- NeurIPS '23 Thomas Hartvigsen, Swami Sankaranarayanan, Hamid Palangi, Yoon Kim, Marzyeh Ghassemi. "Aging with GRACE: Lifelong Model Editing with Discrete Key-Value Adaptors." In *Proceedings of the 35th Conference on Neural Information Processing Systems*.
- ASRU '23 Cheng-I Jeff Lai, Freda Shi, Puyuan Peng, Yoon Kim, Kevin Gimpel, Shiyu Chang, Yung-Sung Chuang, Saurabhchand Bhati, David Cox, David Harwath, Yang Zhang, Karen Livescu, James Glass. "Audio-Visual Neural Syntax Acquisition." In *2023 IEEE Automatic Speech Recognition and Understanding Workshop*.
- ACL '23 Songlin Yang, Roger Levy, Yoon Kim. "Unsupervised Discontinuous Constituency Parsing with Mildly Context-Sensitive Grammars." In *Proceedings of the 2023 Annual Conference of the Association for Computational Linguistics*.
- ACL '23 Lucas Torroba Hennigen, Yoon Kim. "Deriving Language Models from Masked Language Models." In *Proceedings of the 2023 Annual Conference of the Association for Computational Linguistics*.
- ACL '23 Jiaxin Ge, Hongyin Luo, Yoon Kim, James Glass. "Entailment as Robust Self-Learner." In *Proceedings of the 2023 Annual Conference of the Association for Computational Linguistics*.
- ICLR '23 Peihao Wang, Rameswar Panda, Lucas Torroba Hennigen, Philip Greengard, Leonid Karlinsky, Rogerio Feris, David Cox, Zhangyang Wang, Yoon Kim. "Learning to Grow Pretrained Models for Efficient Transformer Training." In *Proceedings of the Eleventh International Conference on Learning Representations*.
- ICLR '23 Zhen Wang, Rameswar Panda, Leonid Karlinsky, Rogerio Feris, Huan Sun, Yoon Kim. "Multitask Prompt Tuning Enables Parameter-efficient Transfer Learning." In *Proceedings of the Eleventh International Conference on Learning Representations*.
- ICLR '23 Han Guo, Philip Greengard, Hongyi Wang, Andrew Gelman, Yoon Kim, Eric Xing. "Federated Learning as Variational Inference: A Scalable Expectation Propagation Approach." In *Proceedings of the Eleventh International Conference on Learning Representations*.
- EMNLP '22 Bailin Wang, Ivan Titov, Jacob Andreas, Yoon Kim. "Hierarchical Phrase-based Sequence-to-Sequence Learning." In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*.
- EMNLP '22 Monica Agrawal, Stefan Hegselmann, Hunter Lang, Yoon Kim, David Sontag. "Large Language Models are Zero-Shot Clinical Information Extractors." In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*.
- ICML '22 Hunter Lang, Monica Agrawal, Yoon Kim, David Sontag. "Co-training Improves Prompt-based Learning with Large Language Models" In *Proceedings of the 39th International Conference on Machine Learning*.

- NAACL '22 Yung-Sung Chuang, Rumen Dangovski, Hongyin Luo, Yang Zhang, Shiyu Chang, Marin Soljagic, Shang-Wen Li, Wen-tau Yih, Yoon Kim, James Glass. "DiffCSE: Difference-based Contrastive Learning for Sentence Embeddings." In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics*.
- NAACL '22 Andrew Drozdov, Jiawei Zhou, Radu Florian, Andrew McCallum, Tahira Naseem, Yoon Kim, Ramon Fernandez Astudillo. "Inducing and Using Alignments for Transition-based AMR Parsing." In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics*.
- CVPR '22 Yi Li, Rameswar Panda, Yoon Kim, Chun-Fu Chen, Rogerio Feris, David Cox, Nuno Vasconcelos. "VALHALLA: Visual Hallucination for Machine Translation." In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.
- ACL '22 Jiabao Ji, Yoon Kim, James Glass, Tianxing He. "Controlling the Focus of Pretrained Language Generation Models." In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Findings)*.
- NeurIPS '21 Yoon Kim. "Sequence-to-Sequence Learning with Latent Neural Grammars." In *Proceedings of the 35th Conference on Neural Information Processing Systems*.
- ACL '21 Demi Guo, Alexander M. Rush, Yoon Kim. "Parameter-Efficient Transfer Learning with Diff Pruning." In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*.
- VIS '21 Shahid Latif, Zheng Zhou, Yoon Kim, Fabian Beck, Nam Wook Kim. "Kori: Interactive Synthesis of Text and Charts in Data Documents." In *Proceedings of the 2021 IEEE Visualization conference*.
- MICCAI '21 Stanislav Lukyanenko, Won-Dong Jang, Donglai Wei, Robbert Struyven, Yoon Kim, Brian Leahy, Helen Yang, Alexander Rush, Dalit Ben-Yosef, Daniel Needleman, Hanspeter Pfister. "Developmental Stage Classification of Embryos Using Two-Stream Neural Network with Linear-Chain Conditional Random Field". In *Proceedings of the 24th International Conference on Medical Image Computing and Computer Assisted Intervention*.
- EMNLP '20 Demi Guo, Yoon Kim, Alexander M. Rush. "Sequence-Level Mixed Sample Data Augmentation." In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*.
- ICML '20 Rares-Darius Buhai, Yoni Halpern, Yoon Kim, Andrej Risteski, David Sontag. "Empirical Study of the Benefits of Overparameterization in Learning Latent Variable Models." In *Proceedings of the 37th International Conference on Machine Learning*.
- ICML '20 Jonathan Mamou, Hang Le, Miguel Del Rio, Cory Stephenson, Hanlin Tang, Yoon Kim, SueYeon Chung. "Emergence of Separable Manifolds in Deep Language Representations." In *Proceedings of the 37th International Conference on Machine Learning*.
- NeurIPS '19 Sam Wiseman, Yoon Kim. "Amortized Bethe Free Energy Minimization for Learning MRFs." In *Proceedings of the 33rd Conference on Neural Information Processing Systems*.
- ACL '19 Yoon Kim, Chris Dyer, Alexander M. Rush. "Compound Probabilistic Context-Free Grammars for Grammar Induction." In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*.
- NAACL '19 Yoon Kim, Alexander M. Rush, Lei Yu, Adhiguna Kuncoro, Chris Dyer, Gábor Melis. "Unsupervised Recurrent Neural Network Grammars." In *Proceedings of the 17th Annual Conference of the North American Chapter of the Association for Computational Linguistics*.

- AISTATS '19 Adji B. Dieng, Yoon Kim, Alexander M. Rush, David M. Blei. "Avoiding Latent Variable Collapse with Generative Skip Models." In *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics*.
- NeurIPS '18 Yuntian Deng, Yoon Kim, Justin Chiu, Demi Guo, Alexander M. Rush. "Latent Alignment and Variational Attention." In *Proceedings of the 32nd Conference on Neural Information Processing Systems*.
- ICML '18 Yoon Kim, Sam Wiseman, Andrew C. Miller, David Sontag, Alexander M. Rush. "Semi-Amortized Variational Autoencoders." In *Proceedings of the 35th International Conference on Machine Learning*.
- ICML '18 Jake Zhao, Yoon Kim, Kelly Zhang, Alexander M. Rush, Yann LeCun. "Adversarially Regularized Autoencoders." In *Proceedings of the 35th International Conference on Machine Learning*.
- EMNLP '17 Allen Schmalz, Yoon Kim, Alexander M. Rush, Stuart M. Shieber. "Adapting Sequence Models for Sentence Correction." In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*.
- ACL '17 Guillaume Klein, Yoon Kim, Yuntian Deng, Jean Senellart, Alexander M. Rush. "OpenNMT: Open-Source Toolkit for Neural Machine Translation." In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (System Demonstrations)*.
- ICLR '17 Yoon Kim, Carl Denton, Luong Hoang, Alexander M. Rush. "Structured Attention Networks." In *Proceedings of the 5th International Conference on Learning Representations*.
- EMNLP '16 Yoon Kim, Alexander M. Rush. "Sequence-Level Knowledge Distillation." In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*.
- AAAI '16 Yoon Kim, Yacine Jernite, David Sontag, Alexander M. Rush. "Character-Aware Neural Language Models." In *Proceedings of the Thirtieth AAAI Conference on Artificial Intelligence*.
- EMNLP '14 Yoon Kim. "Convolutional Neural Networks for Sentence Classification." In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing*.

Workshop & Other Publications

- 2021 Matteo Alleman, Jonathan Mamou, Miguel A Del Rio, Hanlin Tang, Yoon Kim, SueYeon Chung. "Syntactic Perturbations Reveal Representational Correlates of Hierarchical Phrase Structure in Pretrained Language Models." In *Proceedings of the 6th Workshop on Representation Learning for NLP*.
- 2018 Yoon Kim, Sam Wiseman, Alexander M. Rush. "A Tutorial on Deep Latent Variable Models of Natural Language." Tutorial at EMNLP.
- 2018 Gabriel Grand, Aron Szanto, Yoon Kim, Alexander M. Rush. "On the Flip Side: Identifying Counterexamples in Visual Question Answering." In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (Deep Learning Day)*.
- 2016 Allen Schmalz, Yoon Kim, Alexander M. Rush, Stuart M. Shieber. "Sentence-Level Grammatical Error Identification as Sequence-to-Sequence Correction." In *Proceedings of the 11th Workshop on Innovative Use of NLP for Building Educational Applications*.
- 2014 Yoon Kim, Yi-I Chiu, Kentaro Hanaki, Darshan Hegde, Slav Petrov. "Temporal Analysis of Language through Neural Language Models." In *Proceedings of the ACL 2014 Workshop on Language Technologies and Computational Social Science*.

- 2014 Yoon Kim, Owen Zhang. “Credibility Adjusted Term Frequency: A Supervised Term Weighting Scheme for Sentiment Analysis and Text Classification.” In *Proceedings of the 5th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media*.

Professional Activities

- 2024 Simons Institute for the Theory of Computing Workshop Co-organizer Emerging Generalization Settings
 2022 EMNLP Workshop Co-organizer: Unimodal and Multimodal Induction of Linguistic Structures
 2019 ICLR Workshop Co-organizer: Deep Generative Models for Highly Structured Data.
 2018 EMNLP Tutorial: Deep Latent Variable Models of Natural Language.

Area Chair: NeurIPS (2024), ICLR (2023, 2024, 2025), EMNLP (2021, 2022, 2023), COLING (2022), NAACL (2021, 2024), EMNLP (2021)

Reviewing: AAAI (2020), ACL (2017, 2018, 2019, 2020), CoNLL (2018, 2019), EMNLP (2018, 2019, 2020), ICLR (2019, 2020, 2021, 2022), ICML (2019), NAACL (2019), NeurIPS (2018, 2019, 2020), UAI (2018, 2019)

Talks

- Oct 2024 Aspen Meeting on Foundations Models
 Oct 2024 University of Washington, NLP Seminar
 Aug 2024 KAIST, NLP Workshop
 Aug 2024 Seoul National University, Data Science Colloquium
 Mar 2024 University of Oregon, AI Seminar
 Mar 2024 KAIST, Graduate School of AI Colloquium
 Nov 2023 Cornell Tech, Learning Machines Seminar
 Nov 2023 New York University, Text as Data Seminar
 Oct 2023 University of Pennsylvania, NLP Seminar
 Jul 2022 KAIST, ERC AI Seminar
 May 2022 University of Minnesota, NLP Seminar
 Apr 2022 University of Edinburgh, ILCC Seminar
 Apr 2022 Georgia Tech, NLP Seminar
 Apr 2022 Harvard Linguistics Department, Universals Workshop
 Oct 2020 Stanford, NLP Seminar
 Oct 2020 Seoul National University, Data Science Seminar
 Sep 2020 KAIST, CS Colloquium
 Aug 2020 Seoul National University, AI Summer School
 Apr 2020 University of Toronto, CS Colloquium
 Apr 2020 University of Chicago, CS Colloquium
 Mar 2020 MIT, EECS Colloquium
 Feb 2020 Carnegie Mellon University, CS Colloquium
 Feb 2020 Harvard University, CS Colloquium
 Feb 2020 New York University, CS Colloquium
 Feb 2020 Columbia University, CS Colloquium
 Jan 2020 Toyota Technological Institute at Chicago, Colloquium
 Sep 2019 MIT, Computational Psycholinguistics Lab
 Sep 2019 UMass Amherst, Machine Learning and Friends Lunch
 Jan 2018 KAIST, Computational NeuroSystems Lab
 Jan 2018 Seoul National University, Vision and Learning Lab
 May 2017 OpenAI
 Apr 2017 Boston Machine Learning Meetup
 Apr 2016 MIT, Spoken Language Systems Group
 Dec 2015 MIT, NLP Group
 Oct 2014 City University of New York, NLP Seminar